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Identification of the citrus GRF gene family and its expression in fruit peel thickening mediated by gibberellin

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Abstract

Background Growth-regulating factors (GRFs) play a crucial role in plant growth and development, particularly in cell division and expansion. Citrus fruit cracking, a prevalent issue, adversely impacts both yield and fruit quality. Gibberellins (GAs) are known to ameliorate citrus fruit cracking by inducing thicker peel formation, which is attributed to cell division and expansion. However, the mechanistic link between gibberellins and citrus peel thickening, and whether this process is mediated by *GRF* genes regulation, has not been definitively established.

Results In this study, 8 *CsGRFs* (*Citrus sinensis*), 11 *CcGRFs* (*Citrus clementina*), and 8 *CgGRFs* (*Citrus grandis*) were identified from the citrus genome which divided into six clusters, with the genes of the same cluster sharing similar gene structures. *Cis*-elements analysis revealed that the promoter regions of *GRF* genes contained numerous hormone-responsive elements. Tissue expression profiles showed that *CsGRF* genes had higher expression levels in young tissues, including early fruit tissues, one-year-leaf, ovules, and root tips. RNA-seq and qPCR analyses revealed that the expression levels of *CsGRF3*, 4, 7, and 8 were significantly regulated in response to GA₃ treatment. Notably, *CsGRF8* was the most significantly induced by GA₃ and highly expressed in the early stages of peel development. These findings indicate gibberellins may exert regulatory effects on peel development through the induction of *CsGRF* genes.

Conclusion This study systematically analyzed the characteristics of the citrus GRF gene family, as well as the changes in citrus peel thickness and the expression patterns of *CsGRF* genes under gibberellin treatment. These findings provide valuable insights for advancing research on the role of *CsGRF* genes in regulating citrus peel development, which could help reduce the occurrence of fruit cracking.

Clinical trial number Not applicable.

Keywords GRF, Citrus, Gibberellins, Peel thickness, Fruit cracking

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Background

Peel thickness is an important factor affecting the quality of citrus fruits, including excessively thin or thick peels. Excessively thin peel development easily leads to preharvest fruit cracking and reduced storage quality [1–3]. Conversely, excessively thick peel development is often associated with a decline in internal quality, such as reduced total soluble solids (TSS) content and weakened flavor [4, 5]. These abnormalities in peel thickness ultimately result in an overall decline in fruit quality. Therefore, unraveling the molecular mechanisms that regulate peel thickness is of paramount importance for advancing citrus fruit quality.

Gibberellins, a class of plant growth regulators, are integral to a multitude of plant developmental processes including leaf expansion [6], stem elongation [7], and root growth [8]. Despite this, studies focusing on their impact during fruit development, and specifically on peel thickness modulation, are comparatively scant. Notably, gibberellin application has been shown to enhance peel thickness in *Citrus unshiu*, which is associated with the formation of rough fruit [4]. Early maturing ponkan varieties exhibit an increase in peel thickness in response to gibberellin treatment, which correlates with a decreased incidence of fruit cracking, without compromising fruit quality [9]. In tomatoes, silencing the *SIPRE2* gene upregulates GA biosynthesis and catabolism genes, resulting in a significant reduction in peel cell size and overall peel thickness [10]. Conversely, interfering with the expression of the *SIDELLA* protein, a key repressor in the GA signaling pathway, increases the length of peel cells, particularly in the mesocarp, where cell length nearly doubles [11]. While the influence of gibberellin on peel cell development and thickness has been established, the molecular mechanisms underlying these processes warrant further investigation.

GRFs are plant-specific transcription factors that have been identified across many species, including *Arabidopsis thaliana* [12], *Zea mays* [13], *Solanum lycopersicum* [14], *Glycine max* [15], *Oryza sativa* [16], *Nicotiana tabacum* [17], *Arachis hypogaea* [18], *Brassica napus* [19]. *GRF* genes are extensively studied for their roles in leaf morphology [20, 21], grain size regulation [22, 23], stress responses [24], and hormone responses [25, 26]. *GRF* genes were first discovered in rice, where they play a crucial role in regulating gibberellin-induced stem elongation and leaf development [6, 27]. Overexpression of the *OsGRF7* gene upregulates the GA signaling negative regulators *SLR1* and *SLR1L*, ultimately leading to altered leaf morphology and reduced plant height [28]. In maize, overexpression of the *ZmGRF10* gene results in a reduced rate of cell division, leading to smaller leaf size and shorter plant height [29]. In *Arabidopsis*, mutants of *GRF1*, 2, and 3 result in an

enlargement of the meristematic region in the root tips, whereas overexpression of the *AtGRF2* and *AtGRF3* exhibit the opposite phenotype [30]. Several studies have established that miR396 can directly target and regulate *GRFs* [31–33]. In tomatoes, the disruption of miR396 leads to a significant upregulation of *SIGRF1/2/3/4/5*, resulting in the enlargement and thickening of fruit peel cells, and further observations revealed that overexpression of *SIGRF2/3/4/5* also caused an increase in the thickness of the fruit pericarp, particularly with *SIGRF4* [14, 34].

The N-terminal region of GRF proteins contains two highly conserved domains: the QLQ (Gln, Leu, Gln) and WRC (Trp, Arg, Cys) domains [12, 31]. The QLQ domain interacts with the SNH domain (SYT N-terminal homology domain) of GIF (GRF-interacting factor) proteins, thereby facilitating its transcriptional activation function [27, 35]. The WRC domain contains two distinctive features, namely a putative nuclear localization signal and a zinc-finger motif that consists of the conserved spacing of three Cys and one His residue [12, 27]. Unlike the highly conserved N-terminal region, the C-terminal region of GRF proteins, characterized by less conserved motifs such as TQL (Thr, Gln, Leu) and FFD (Phe, Phe, Asp), exhibits significant variability in length and amino acid composition and have a role in transactivation [13, 29, 36]. These unconservative structural features in C-terminal implying that *GRF* genes may possess diverse biological functions.

Fruit cracking presents a considerable challenge in the citrus industry, adversely affecting both yield and fruit quality. Understanding whether *GRF* genes regulate citrus peel thickness in response to gibberellin could help mitigate fruit cracking. This study performed a comprehensive genome-wide identification of the *GRF* gene family in *Citrus sinensis*, *Citrus clementina*, and *Citrus grandis*. It analyzed their evolutionary relationships, gene structures, cis-elements, chromosomal locations, duplication events, and expression patterns across various citrus tissues. Additionally, the study investigated the effects of different concentrations of gibberellins on peel thickness and *CsGRFs* transcription levels. The aim is to identify *CsGRF* genes affecting citrus peel thickness, providing a foundation for improving citrus quality.

Results

Genome-wide identification of GRF gene family in *C. sinensis*, *C. clementina* and *C. grandis*

To comprehensively identify potential *GRF* genes in the citrus genomes of *C. sinensis*, *C. clementina*, and *C. grandis*, we utilized the HMM model based on the conserved QLQ and WRC domains. This screening identified 8 *GRF* genes in *C. sinensis*, 11 in *C.*

clementina, and 8 in *C. grandis*. We conducted the basic characteristics for these GRFs, encompassing aspects such as chromosome localization, number of introns, protein length, isoelectric point, molecular weight, and subcellular localization (Table S1).

In CsGRFs, CcGRFs, and CgGRFs, intron numbers ranged 1–3, 1–3, and 2–3, respectively. Protein lengths spanned 232 aa (CsGRF5)–600 aa (CsGRF2), 232 aa (CcGRF3)–600 aa (CcGRF10), and 312 aa (CgGRF6)–600 aa (CgGRF1), respectively. Isoelectric points (pI) ranged 5.48 (CsGRF4)–9.75 (CsGRF5), 5.83 (CcGRF4)–9.09 (CcGRF8), and 5.98 (CgGRF5)–9.03 (CgGRF3), respectively. Molecular weights (Mw) ranged 25.26 KDa (CsGRF5)–65.05 KDa (CsGRF2), 25.40 KDa (CcGRF3)–65.17 KDa (CcGRF10), and 34.98 KDa (CgGRF6)–65.05 KDa (CgGRF1), respectively. All predicted subcellular localizations indicated nuclear localization.

Phylogenetic analysis of GRF gene family between *C. sinensis*, *C. clementina*, *C. grandis*, and *A. thaliana*

Phylogenetic analysis based on protein sequence data from *C. sinensis*, *C. clementina*, *C. grandis*, and *A. thaliana* clustered the 36 GRFs from these four species into six groups (I–VI) (Fig. 1). Group III contained the most members (9), followed by Groups IV and V (7 each), Groups I and II (5 each), and Group VI, which had the fewest members (3) and lacked CgGRF members. Interestingly, group V included only GRFs from the

Citrus genus, suggesting a potential genus-specific uniqueness.

Gene structures, conserved motifs, and conserved domains analysis of GRF gene family in *C. sinensis*, *C. clementina*, and *C. grandis*

To explore evolutionary relationships and gene function, we analyzed the gene structures and conserved motifs in *CsGRFs*, *CcGRFs*, and *CgGRFs*. The gene structure analysis revealed that *GRFs* contained 2–4 exons (Fig. 2). The WRC and QLQ domains were consistently located at the N-terminus of the GRF proteins. Through motif analysis, 10 conserved motifs were identified, with motif 1 and motif 2 present in all GRF proteins, indicating that this structural domain was highly conserved within the GRF gene family. Group IV had the fewest motifs, containing only motifs 1, 2, and 3, whereas Group V had the most, with 7–8 motifs. Similar motif patterns were observed within the same group, suggesting analogous gene functions, while variations in motif order and number across different groups might contribute to functional diversities.

Cis-elements in the promoters of GRF gene family in *C. sinensis*, *C. clementina*, and *C. grandis*

Cis-elements in the promoter region, crucial for gene regulation, were identified in the *GRF* genes of *C. sinensis*, *C. clementina*, and *C. grandis*. We screened for hormone-responsive and stress-responsive *cis*-elements, including auxin-responsive elements (TGA-element, AuxRE, TGA-box, AuxRR-core), gibberellin-responsive elements (TATC-box, P-box, GARE-motif), MeJA-responsive elements (TGACG-motif, CGTCA-motif), abscisic acid-responsive elements (ABRE), ethylene-responsive elements, salicylic acid-responsive elements (TCA-element); defense and stress-responsive elements (TC-rich repeats), wound-responsive elements (WUN-motif), MYB binding sites involved in drought-inducibility (MBS), low temperature-responsive elements (LTR), and anaerobic induction elements (ARE), totaling 16 types of *cis*-elements (Fig. 3).

MeJA-responsive elements (TGACG-motif or CGTCA-motif) were the most abundant hormone-responsive elements in the promoter regions of 27 *GRF* genes, with 20 *GRF* gene promoters containing this element, particularly abundant in Group V. Additionally, abscisic acid-responsive (ABRE), auxin-responsive (TGA-element, AuxRE, TGA-box, or AuxRR-core), and gibberellin-responsive (TATC-box, P-box, or GARE-motif) elements were identified in 16, 13, and 13 gene promoters, respectively. In stress responses, anaerobic induction (ARE) elements were found in 21 gene promoters, while drought-inducibility (MBS) and low temperature-responsive (LTR) elements were found in

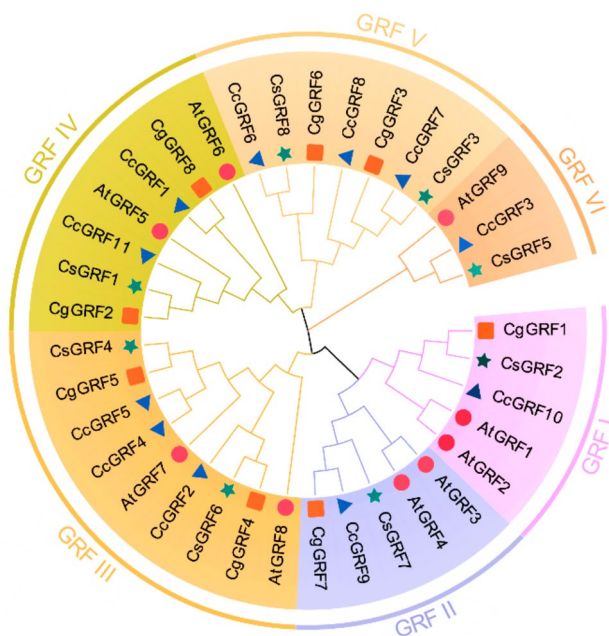


Fig. 1 Phylogenetic tree of GRF proteins in *C. sinensis*, *C. clementina*, *C. grandis* and *A. thaliana*. The phylogenetic tree was derived using MEGA11 software by neighbor-joining method with 1000 bootstrap replicates. Species abbreviations are listed as follows. Cs: *C. sinensis*; Cc: *C. clementina*; Cg: *C. grandis*; At: *A. thaliana*

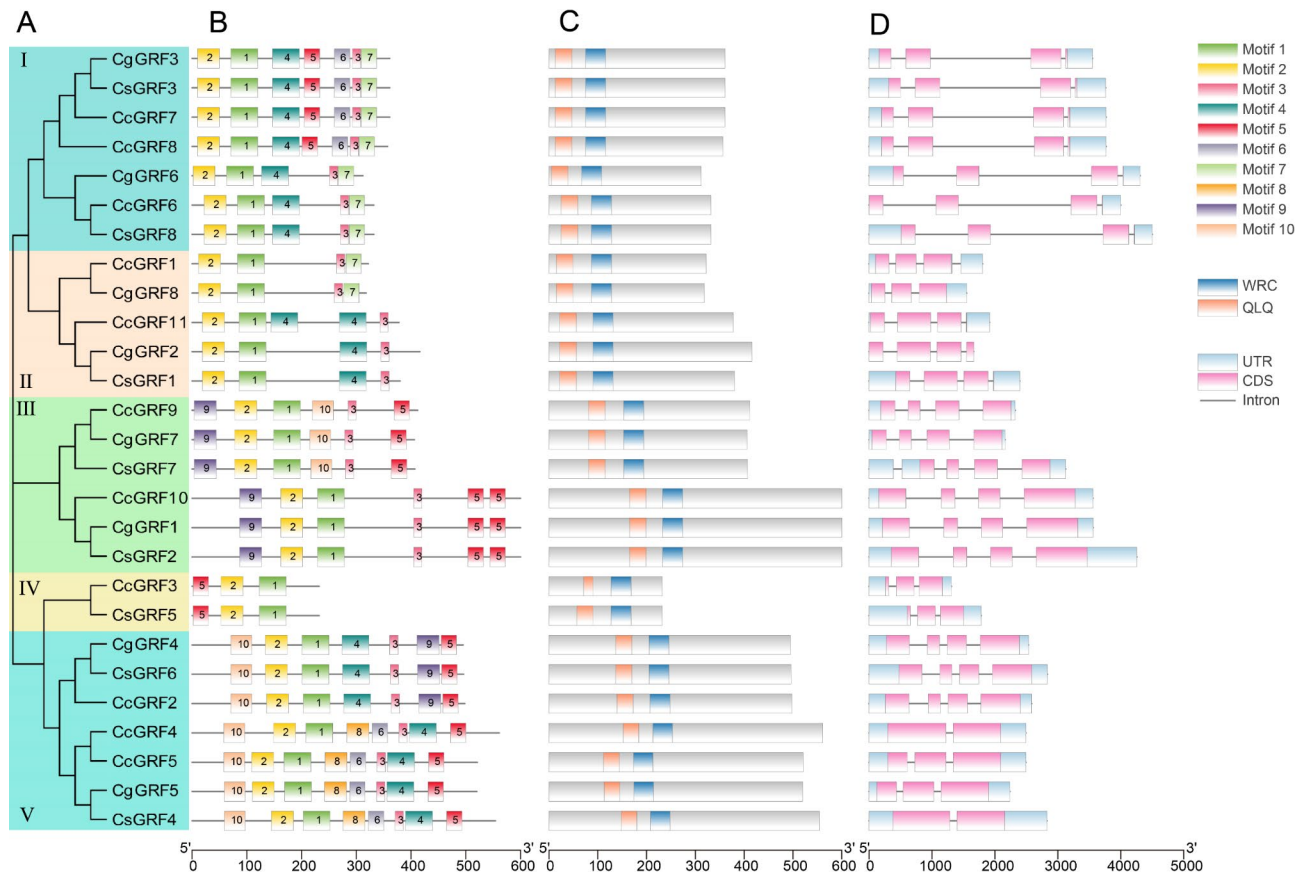


Fig. 2 Phylogenetic relationships, conserved motif structures, and exon-intron structures of CsGRFs, CcGRFs, and CgGRFs. **A** The phylogenetic tree was constructed using MEGA11 software by neighbor-joining method with 1000 bootstrap replicates. **B** Distribution of conserved motifs in CsGRFs, CcGRFs and CgGRFs. The number of motifs 1–10 was displayed in different colored boxes. **C** Conserved domain of CsGRFs, CcGRFs and CgGRFs showing distribution of WRC and QLQ domain. **D** Structure of CsGRF, CcGRF and CgGRF genes showing distribution of UTR regions (blue bars), CDS (pink bars) and introns (black lines)

12 and 11 gene promoters, respectively. These findings suggest that the GRF gene family plays significant roles in various hormonal and stress responses.

Duplication events of GRF gene family

To further investigate the evolutionary mechanisms of GRF gene family, we analyzed the chromosomal localization and tandem/segmental duplication events of *CsGRFs*, *CcGRFs*, and *CgGRFs*. This study showed that *CsGRFs*, *CcGRFs*, and *CgGRFs* were distributed on chromosomes chr1, chr3, chr5, and chr6 simultaneously (Fig. S1).

No tandem duplication events were found in *CsGRFs*, *CcGRFs*, and *CgGRFs*. *C. sinensis* and *C. clementina* contained a pair of segmental duplicated genes (*CsGRF4/6*, *CcGRF2/6*, respectively), while *C. grandis* did not contain any (Fig. 4A). These results imply that segmental duplication predominantly facilitate the expansion of *GRF* genes.

Comparative syntenic maps constructed among *C. sinensis*, *C. clementina*, *C. grandis*, and *A.*

thaliana revealed 12, 8, and 8 syntenic pairs between *CsGRFs/CcGRFs*, *CsGRFs/CgGRFs* and *CcGRFs/CgGRFs*, respectively. 9, 10, and 7 syntenic pairs between *CsGRFs/CcGRFs/CgGRFs* and *AtGRFs*, respectively (Fig. 4B, Table S2). Notably, *CgGRF4* contained 6 collinear gene pairs, while *CsGRF4/6* and *CcGRF2/4* each contained 5 collinear gene pairs. This suggests that these *GRF* genes may play a crucial role in the evolution of *GRF* gene family.

Influence of gibberellins on peel thickness

To evaluate the influence of exogenous gibberellin (GA_3) on peel thickness, different GA_3 concentrations (25, 50, and 75 mg/L) were applied to ‘Zaomi’ (*C. reticulata* Blanco) during the early fruit expansion stage, and peel thickness was measured in the late expansion stage and maturity stage, respectively. The results showed that GA_3 application was associated with an enhancement in peel roughness and an expansion in fruit size, as well as fruit coloring (Fig. S2–S4). After treatment with 25/50/75 mg/L of GA_3 , the peel thickness increased from

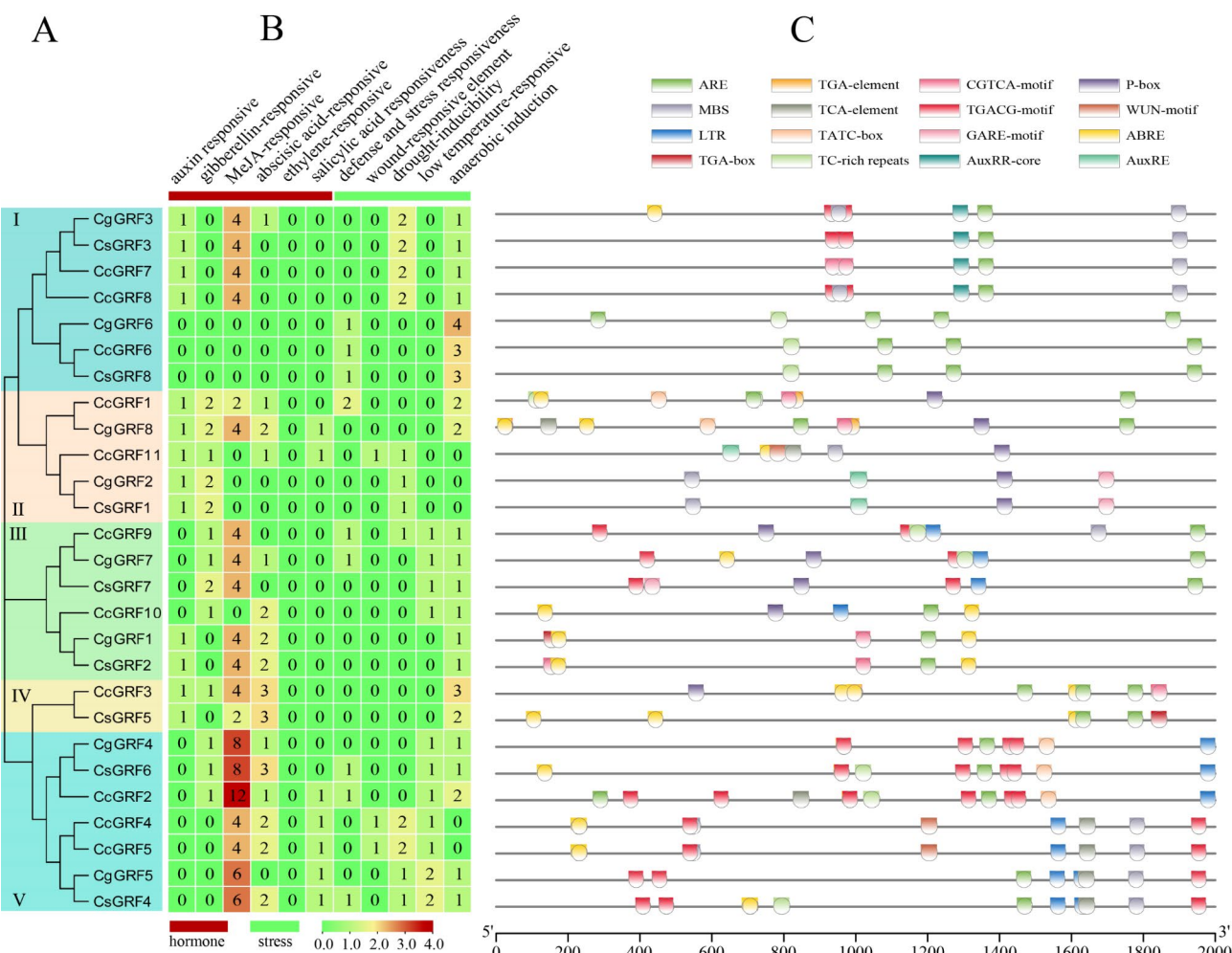


Fig. 3 Cis-element analysis of the *CsGRF*, *CcGRF*, and *CgGRF* genes in citrus. **A** The phylogenetic tree was constructed using MEGA11 software by neighbor-joining method with 1000 bootstrap replicates. **B** Category and quantity of 11 hormone response and stress response elements. **C** Detailed types and locations of 11 hormone response and stress response elements

1.76 mm to 2.03, 2.25, 2.25 mm in the late expansion stage, and from 2.28 mm to 2.64, 2.59, 2.57 mm in the maturity stage, respectively (Fig. 5). The results indicate that different concentrations of GA₃ promote an increase in peel thickness.

Tissue expression profiles of *CsGRF* genes

Since *Citrus sinensis* v3.0 possesses relatively comprehensive genomic information, and *CsGRFs* exhibit high sequence homology (Table S4) with both *CcGRFs* and *CgGRFs*, *CsGRFs* were chosen for subsequent analysis.

The expression profiles of *CsGRF* genes were analyzed across different citrus tissues, including different developmental stages of leaves, ovules, and roots. The expression level of *CsGRF* genes were higher in ovules and young leaves, while lower in roots and mature leaves. This indicates that *CsGRF* genes exhibits spatiotemporally expression specificity pattern, which

higher expression in ovules suggest that its potential role in fruit development (Fig. 6A).

To further explore the role of *CsGRF* genes in fruit development, we analyzed the expression levels of *CsGRF* genes in different development stages of the epicarp, albedo, juice sacs, and segment membrane in *C. sinensis* fruits (Fig. 6B). *CsGRF3* and *CsGRF7* showed high expression levels in all tissues, while *CsGRF1* and *CsGRF6* were barely expressed across all tissue. *CsGRF2*, 4, 5, and 8 were highly expressed at the early stages of different fruit tissues, with their expression gradually decreasing during fruit maturation. Interestingly, *CsGRF2* and *CsGRF8* exhibited higher expression levels in the epicarp during the early stages of development, suggesting a potential involvement in the progression of peel development.

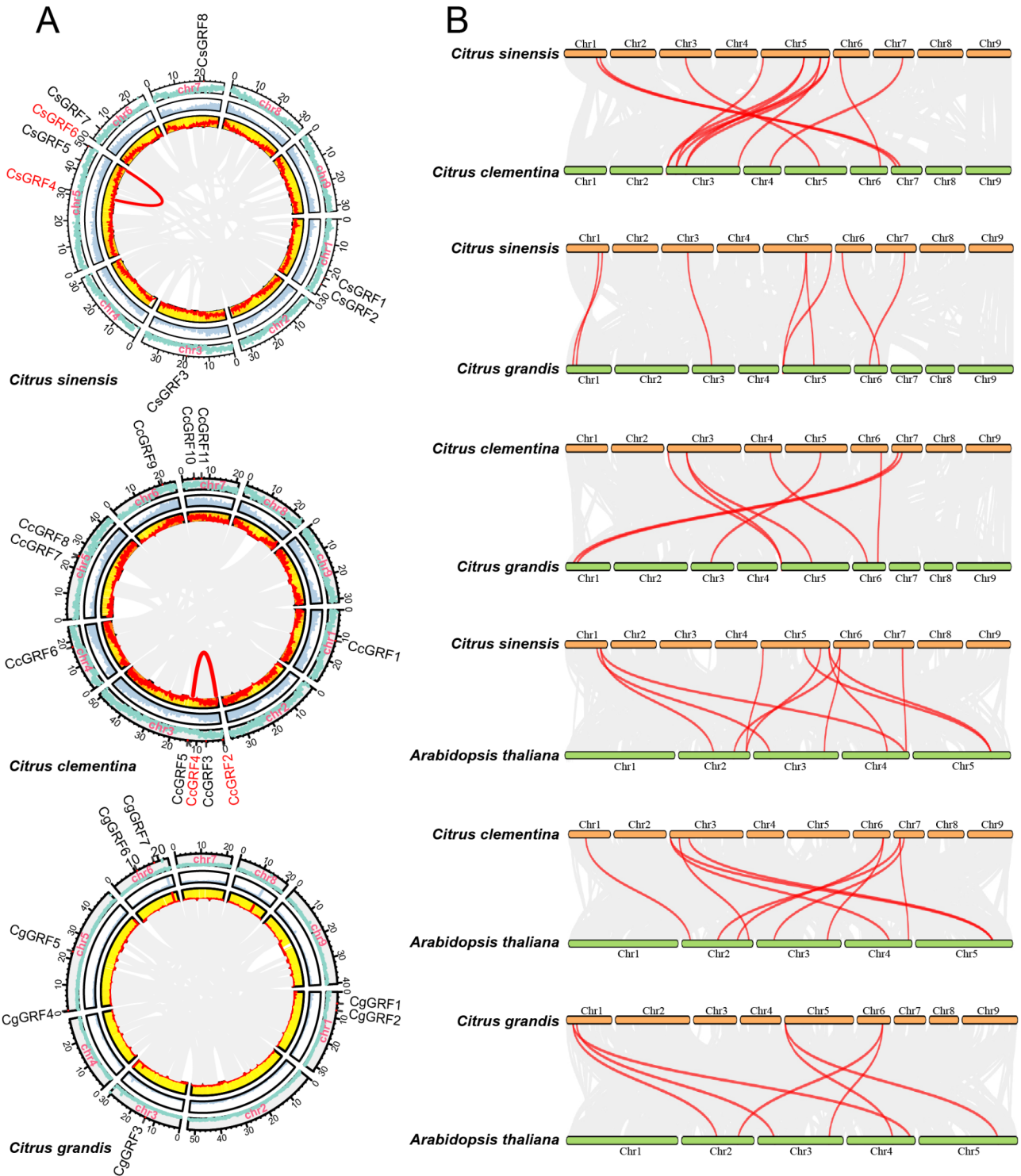


Fig. 4 Intrasppecific and interspecific collinearity analysis citrus GRFs. **A** *CsGRFs*, *CcGRFs* and *CgGRFs* genomic mapping and gene replication analysis. The gray line represents the isogenic pairs, and the red line represents the isogenic pairs of *GRFs*. The peripheral Chr1-Chr9 indicates the chromosome number. The scale marked on the chromosome, indicating chromosome length (Mb). **B** Synteny analyses of *CsGRFs*, *CcGRFs*, *CgGRFs* and *AtGRFs*. Chromosome numbers are marked at the top or bottom of each chromosome. The gray line indicates the isogenic pairs, and the red line between chromosomes indicates the isogenic pairs of *GRFs*

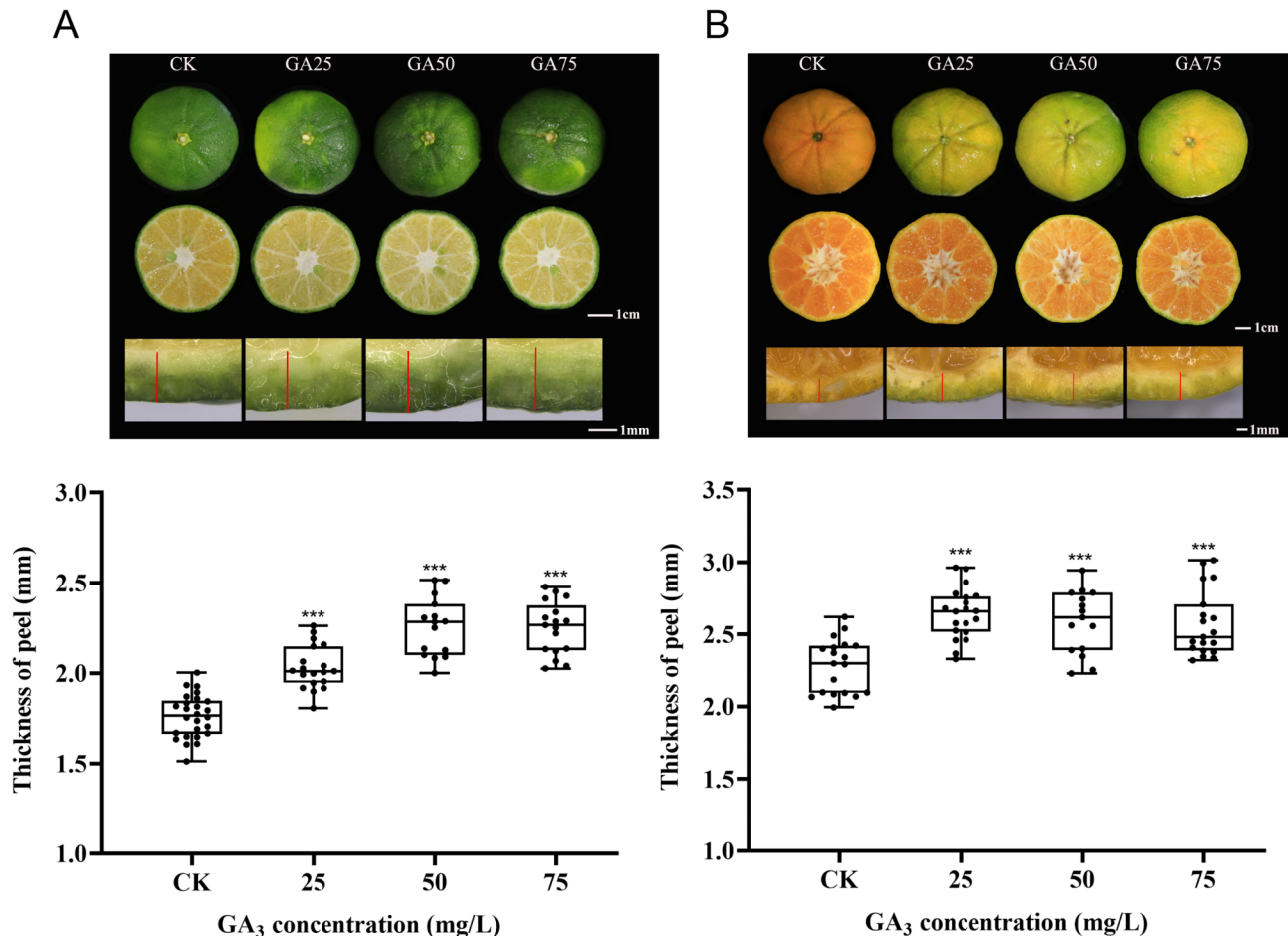


Fig. 5 Peel thickness in 'Zaomi' under different GA_3 concentration. **A** The late expansion stage. **B** The maturation stage. The circular dots on the box plot represent the data for the peel thickness of individual fruits. Statistical analysis of peel thickness after GA_3 treatment. Error bars, mean SD. Data were compared by two-tailed Student's t-test, * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$

RNA-seq and qPCR analyses of *CsGRF* genes expression under GA_3 treatment

To explore whether *CsGRFs* respond to the regulation of GA_3 , RNA-seq analyses of peel samples were performed by different concentrations of GA_3 treatment to 'Zaomi' (Fig. 6C). Results demonstrated that the expression levels of *CsGRF3*, 4, 6, and 8 significantly increased post- GA_3 treatment, particularly *CsGRF8*; whereas *CsGRF7* expression decreased. *CsGRF1*, 2 and 5 expression levels were almost undetectable. These findings indicated *CsGRF* genes were regulated by GA_3 , which suggests that GA_3 may regulate fruit peel development by affecting the expression of *GRFs*.

To further analyze the RNA-seq results, qPCR analysis was performed to assess the expression levels of *CsGRF* genes in 'Zaomi' under different concentrations of GA_3 treatment (Fig. 7). After treatment with GA_3 , the expression levels of *CsGRF3*, 4, and 8 were significantly upregulated, especially *CsGRF8*. Compared with the control, *CsGRF8* showed increments of 459%, 314%, and 186% under 25, 50, and 75 mg/L GA_3 treatment,

respectively. In contrast, the expression of *CsGRF1*, 2, and 7 were significantly downregulated, with *CsGRF7* showing the most considerable decrease, recording reductions of 40%, 34%, and 78% at corresponding GA_3 treatments. Overall, the expression of *CsGRF3*, 4, 7, and 8 was notably responsive to GA_3 modulation, consistent with the RNA-seq results. The expression profiles of tissue-specificity and GA_3 treatment indicate that *GRF8* is highly probable to play a crucial role in fruit peel development.

Predicting the interaction network of *CsGRFs* protein

GRF-GIF often act as partner proteins to form plant-specific transcription complexes that influence organogenesis through involvement in cellular mechanisms [20, 37, 38]. *CsGRF3*, 4, 7, and 8 were significantly regulated by gibberellins, thus their homologous protein interaction networks in *Arabidopsis* were predicted using the STRING online tool (Fig. 8). The results indicate that *CsGRF3*, 4, 7, and 8 homologs in *Arabidopsis* are predicted to interact with GIF1, 2, and 3.

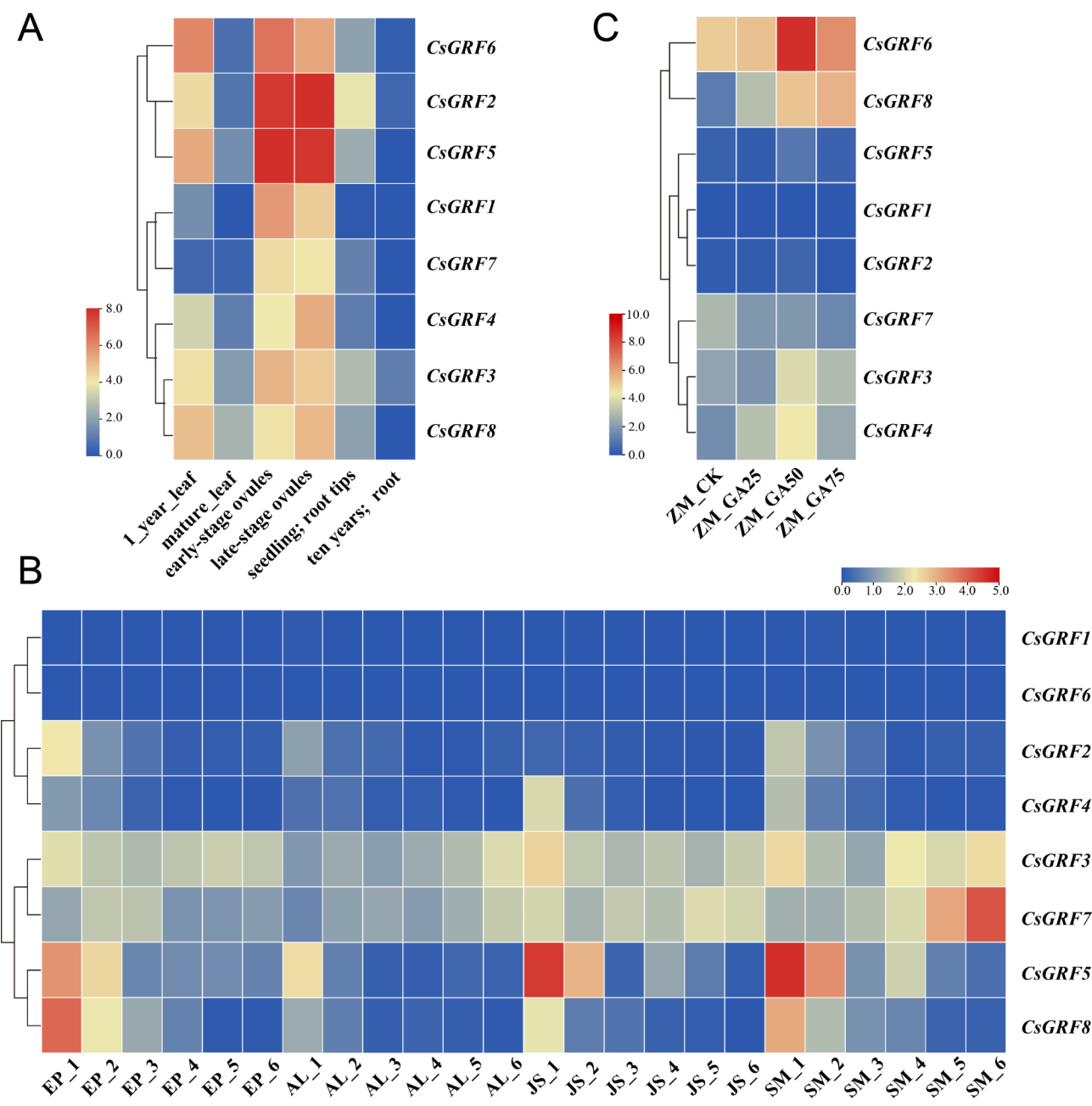


Fig. 6 Expression profiles of *CsGRF1*–*CsGRF8*. **A** The expression pattern of *CsGRFs* in roots, leaves and ovule. The shade of color corresponded to the expression value of *CsGRFs* ($\log_2$ (FPKM + 1)). **B** The expression pattern of *CsGRFs* in various fruit tissues. The color shadows corresponded to *CsGRFs* expression values ($\log_2$ (FPKM + 1)). EP_1–6, AL_1–6, JS_1–6, SM_1–6 (averaged, 3 replicates) represented the epicarp, albedo, juice sac, and segment membrane 50, 80, 120, 150, 180, and 220 days after flowering, respectively. **C** The expression patterns of *CsGRFs* were observed after 25, 50 and 75 mg/L GA_3 treatment of the total peel of ‘Zaomi’. The color shadows corresponded to the FPKM value of *CsGRFs*

Such predicted interactions reinforce the proposed role of the GRF-GIF complexes in the regulatory mechanisms of cellular development.

Discussion

Fruit cracking severely impacts citrus yield and quality, with peel thickness being a critical factor in fruit cracking susceptibility. Previous studies have indicated that cell

size and number play an important role in determining peel thickness [10, 39, 40]. The GRF gene family, unique to plants, has been widely reported to regulate cell size and number, thus contributing to the control of organ size in plants [32, 41, 42]. However, it remains unclear whether citrus *GRF* genes are induced by gibberellin and participate in the formation of fruit peel thickness. In this study, phylogenetic analysis categorized the 36

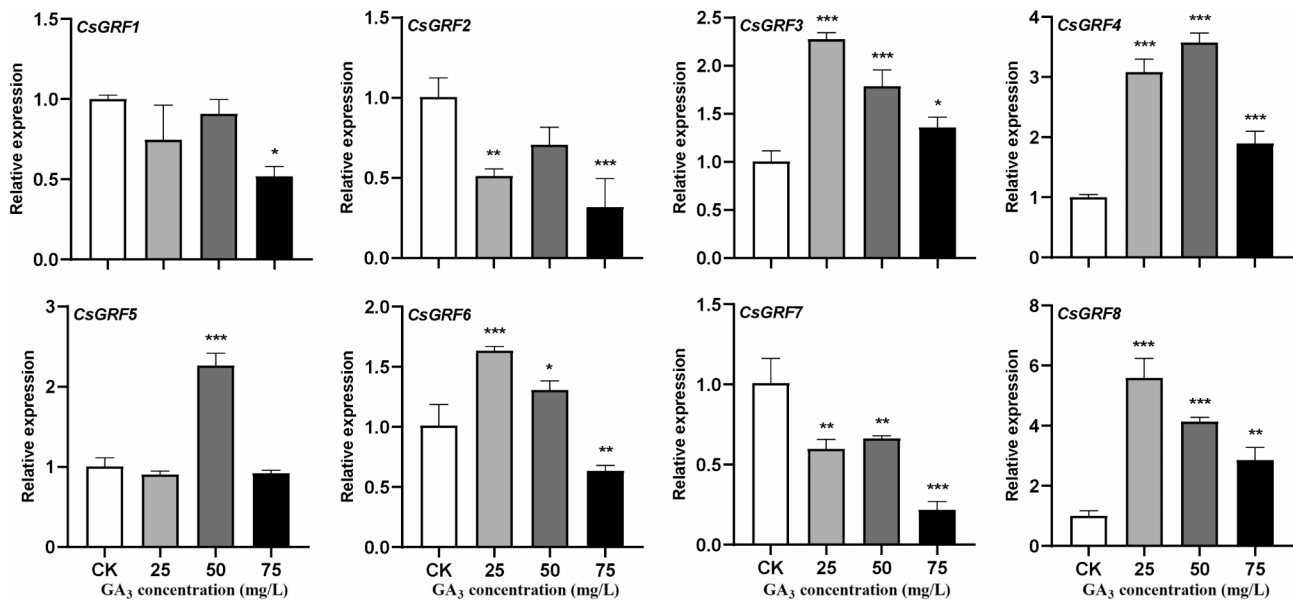


Fig. 7 qPCR analysis of *CsGRF* genes expression in peel treated with different concentrations of gibberellin. Control and gibberellin treatment concentrations were 25, 50 and 75 mg/L, respectively. *Csactin* as internal reference gene; Error bars, mean SD, $n = 3$. Data were compared by two-tailed Student's t-test, * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$

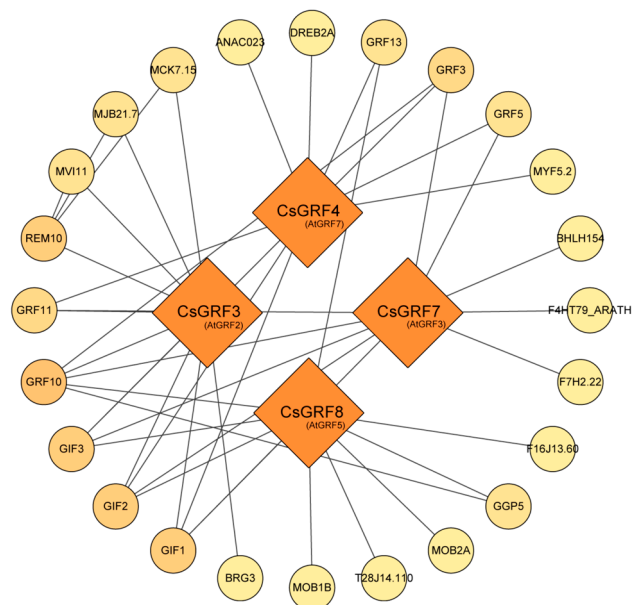


Fig. 8 CsGRFs homologous protein interaction network in *Arabidopsis*. Diamond represents the homologs of *CsGRF3*, 4, 7, and 8 in *Arabidopsis*. Depth of the color indicates the frequency of interaction with other proteins

GRF members from *CsGRFs*, *CcGRFs*, *CgGRFs*, and *AtGRFs* into six distinct groups. Notably, Group V was exclusively composed of GRFs from citrus, hinting at a possible unique evolutionary pathway for this group within the citrus clade. The presence of both motif 1 and motif 2 in all citrus GRFs (Fig. 2) underscores the strong evolutionary conservation characterizing the GRF gene

family. Exon-intron structure and motif distribution analyses were consistent with phylogenetic analysis, revealing high structural similarity among *GRF* genes within the same group but lower similarity between different groups.

Gene duplication provides the raw genetic material for the evolution of novel functions [43]. It is established that the GRF gene family has undergone two notable duplication events: an initial whole-genome triplication event in the shared ancestor of all eudicots, followed by species-specific whole-genome duplications within plants [44]. Citrus, as a core eudicot within the Rutaceae [45], appears to retain a high degree of conservation with limited gene duplication events—often presenting as a single pair or even without duplications. This suggests that *GRF* genes have been preserved throughout the evolutionary trajectory of the citrus lineage. Combining phylogenetic and syntenic analyses based on gene expression can better predict the functions of *GRF* genes in different species. Previous reports indicated that *AtGRF1*, 2, and 3 increase leaf size through cell expansion [12]. In this study, *CsGRF2* was syntenic with *AtGRF1* and *AtGRF2*, while *CsGRF7* and *CsGRF8* were syntenic with *AtGRF2* and *AtGRF3*, respectively, emphasizing the potential functions of *CsGRF2*, 7, and 8 in organ formation similar to *Arabidopsis* by regulating cell expansion.

The structure of citrus fruit comprises the epicarp, albedo, juice sacs and segment membrane, with their formation being closely linked to early differentiation. Citrus peel cells begin dividing before flowering, and

the peel thickens most rapidly during flowering [46]. The GRF gene family is known for its involvement in cell division and expansion particularly in young plant tissues, with their expression typically elevated in the early phases of tissue development and subsequently declining as maturation ensues [47–49]. Consistent with previous findings, *CsGRF* genes also exhibited a specific expression pattern at cell division phase, indicating that they played a major role in the early stages of plant growth and development (Fig. 6A, B). The significant expression of *CsGRF2* and *CsGRF8* within the developing ovule and epicarp coincides with the phase of sharp peel thickening in citrus, indicative of their potential contribution to the development of peel thickness.

Gibberellins are critical hormones in plants, implicated in a wide range of life processes, particularly influencing organ size by modulating both cell division and expansion [50–53], similar to the function of *GRF* genes. The application of exogenous GA_3 can significantly alter the *GRF* gene expression levels [6, 17, 54]. In tomatoes, overexpression of *SIGRF2/3/4/5* leads to different degrees of peel thickening, particularly overexpression of *SIGRF4* increasing it by nearly 50% [34]. In this study, different concentrations of GA_3 increased the peel thickness of ‘Zaomi’ and caused changes in the transcriptional levels of most *CsGRF* genes. This suggests an intrinsic association between *GRF* genes, fruit peel thickness and gibberellins, wherein *GRF* genes potentially play an active role in sustaining or enhancing cellular division and expansion via feedback mechanisms, contributing to organ size regulation. Some studies show that GRF forms a GRF-GIF complex with GIF, which plays a role in regulating organ size due to cell proliferation or expansion, and responds to GA_3 signaling [38, 55, 56]. Whether *CsGRFs*-GIF can form complexes and further play a role in the regulatory network of GA -GRF-GIF-fruit peel thickening requires further experimental verification.

Interestingly, compared to other tissues, *CsGRF8* not only exhibits higher expression levels in the fruit peel but also shows the most significant induction by GA_3 . Meanwhile, *CsGRF8* exhibits the closest evolutionary relationship with *SIGRF4* (Table S3), implying that *CsGRF8* is highly likely to play a similar role in the development of citrus fruit peel thickness. In other words, it seems reasonable to propose that gibberellin promotes peel thickening by inducing the expression of *CsGRF8*. It is noteworthy that the *CsGRF8* promoter lacks gibberellin response elements yet responds significantly to gibberellin. This could be due to gibberellin inducing miR396 to directly target and regulate *CsGRF8* [38, 57]; on the other hand, it could be that the GA -responsive elements are located distally on the gene, regulating the expression of *CsGRF8* through

interactions with transcription factors [58]. These results provide a theoretical basis for comprehending the role of gibberellins in regulating citrus peel thickening by modulating *GRFs* expression, generally further study concentrating on the complexity of molecular regulatory network mediated by hormonal during plant development.

Conclusions

In this study, 8 *CsGRFs*, 11 *CcGRFs*, and 8 *CgGRFs* were identified in the citrus genome. The characteristics, phylogenetic relationships, conserved motifs, gene structures, *cis*-elements, chromosomal distribution, and gene duplication of the 27 citrus *GRF* genes were characterized. Additionally, we investigated the changes in citrus fruit peel thickness following GA_3 treatment and analyzed the expression patterns and transcription levels of the 8 *CsGRF* genes in various citrus tissues and under gibberellin treatment. In fruit tissues, *CsGRF2* and *CsGRF8* exhibited significantly higher expression levels in the epicarp during the early development stages. Further analysis revealed that *CsGRF8* still maintained higher expression levels during the fruit expansion stage and displayed the most significant induction in response to GA_3 . This provides a new direction for further research into how *GRF* genes are associated with GA -mediated fruit peel thickness development.

Methods

Identification and physicochemical property analysis of GRF gene family

The protein sequences and genome annotations for *Citrus sinensis* v3.0, *Citrus clementina* v1.0, and *Citrus grandis* (L.) Osbeck.cv. ‘Wanbaiyou’ v1.0 were retrieved from the citrus genome database (<http://citrus.hzau.edu.cn>). The Hidden Markov Model (HMM) files for the WRC (PF08879) and QLQ (PF08880) domains were obtained from the Pfam database (<http://pfam.xfam.org/>) [59]. Using the HMMER tool in a Linux system, candidate proteins containing both the WRC and QLQ domains were screened to identify *CsGRFs*, *CcGRFs*, and *CgGRFs* ($E\text{-value} \leq 1e^{-5}$). Then, the presumed *GRF* protein sequences were further identified by means of the identification of the QLQ and WRC domains via the SMART (<http://smart.embl-heidelberg.de/>) and InterPro (<https://www.ebi.ac.uk/interpro/search/text/nul1>) databases.

Characterization of the *CsGRFs*, *CcGRFs*, and *CgGRFs* proteins, including amino acid length, isoelectric point, and molecular weight information, was analyzed through the ExPASy ProtParam tool (<https://web.expasy.org/protparam/>). Gene structural analyses were conducted using the Gene Structure View tool within TBtools [60] (<https://github.com/CJ-Chen/TBtools>). Subcellular localization

prediction was performed using the online tool Plant-mPLOC (<http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/>). Furthermore, *CsGRFs*, *CcGRFs*, and *CgGRFs* were renamed based on their chromosome location. The corresponding blast gene mappings for *Citrus sinensis* v2.0, *Citrus sinensis* v1.0, *Citrus clementina* v1.0, and *Citrus sinensis* v3.0 were provided in table (Table S5).

Phylogenetic analysis of the GRF proteins

All GRF protein sequences were aligned using ClustalW within MEGA11 software [61]. Subsequently, an unrooted phylogenetic tree was constructed based on these alignments employing the neighbor-joining method, with 1000 bootstrap replications.

Gene structures, conserved motifs, and conserved domains analysis

The conserved motifs encompassed within the GRF protein sequences were identified through the MEME program (<http://meme-suite.org/tools/meme>). All parameter settings are default, except for the number of motifs set to 10. Conserved domain predictions for GRF proteins were conducted using the Batch CD-Search tool available at the NCBI database (<https://www.ncbi.nlm.nih.gov/Structure/bwrpsb/bwrpsb.cgi>). In instances where the QLQ and WRC domains within GRF proteins were not readily identifiable, the InterPro database (<https://www.ebi.ac.uk/interpro/search/text/null>) was employed as a secondary resource for confirmation. Finally, the gene structures of *CsGRFs*, *CcGRFs*, and *CgGRFs* were visualized with the Gene Structure View tool within TBtools.

Analysis of cis-elements in promoters

The upstream 2000 bp sequence of *CsGRFs*, *CcGRFs*, and *CgGRFs* in the citrus genome was extracted. The online tool PlantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) was utilized for the prediction of cis-regulatory elements within the promoters of GRF genes [62], allowing for the determination of their types, quantities, and associated functions. Visualization of these elements was subsequently achieved using the Basic Biosequence View tool within TBtools.

Chromosomal location and gene duplication analysis

Using the genomic data of *C. sinensis*, *C. clementina*, and *C. grandis*, the chromosomal positions of the GRF genes were mapped using the Gene Location Visualize from GTF/GFF tool within TBtools.

The One StepMCScanX Super Fast tool within TBtools analyzes GRF genes duplication events ($E\text{-value} \leq 1e^{-3}$), and the results were visualized through the Advanced Circos and Dual syteny plot for MCScanX tool within TBtools.

Plant materials and gibberellins treatments

To investigate the effect of GA₃ on the peel thickness development of citrus, the citrus variety ‘Zaomi’ used in this study was sourced from Luxi County, Jishou City, Hunan Province, China. GA₃ at different concentrations (0, 25, 50, and 75 mg/L), with 0.05% Tween-80 included as a surfactant, was applied to the entire tree using a backpack sprayer. GA₃ treatment was applied three times during the early fruit expansion stage (70, 80, and 90 DAF, days after full bloom), with five trees for each treatment. Fruits were harvested during the late expansion stage (110 DAF) and maturation stage (190 DAF).

Measurement of fruit shape indices and peel thickness

Randomly measure the maximum width of over 100 fruits treated with GA₃, and sample based on the fluctuations in the maximum width of the fruits. Measure the maximum width and height of the collected ‘Zaomi’ samples using a vernier scale, and determine the weight of each fruit using an electronic balance. Peel thickness is measured as follows: each fruit was bisected horizontally at its point of maximum width, then peel thickness was gauged at two diametrically opposed sites on the cut surface using a stereo fluorescence microscope. The mean value of these two measurements was recorded as the peel thickness for that particular fruit, with each treatment group containing at least 20 fruits.

Fruit peel RNA sample isolation

Each GA₃ treatment comprised five individual trees. From each tree, we harvested five representative fruits, resulting in a composite of 25 fruits per GA₃ treatment. To ensure sample integrity, we placed the harvested fruits in a portable insulated cooler, alongside ice packs, and transported them to the laboratory on the day of collection. In the laboratory, we meticulously separated the peel (equatorial region) from each fruit. Then, immediately pulverize the fruit peel into powder under liquid nitrogen and store it in a -80°C freezer to preserve the RNA samples for subsequent research (RNA-seq and qPCR).

Tissue expression profiles and RNA-seq analysis

Genome-wide expression data for leaves, root, and ovules tissue of *C. sinensis* were obtained from the citrus genome database (<http://citrus.hzau.edu.cn/>), and detailed data is provided in the table (Table S6). Furthermore, by utilizing the RNA-seq data of *C. sinensis* from Feng [63], the FPKM values for the 8 *CsGRFs* across different tissues (epicarp (EP), albedo (AL), juice sac (JS), and segment membrane (SM)) and developmental stages (50, 80, 120, 155, 180, and 220 DAF). Create a heatmap for these data using the HeatMap tool within TBtools.

The RNA-seq data for ‘Zaomi’ treated with GA₃ can be publicly accessed (BioProject: PRJNA1215682) in the NCBI database. The specific information is as follows: Total RNA from the peel of ‘Zaomi’, collected at the late fruit expansion stage (110 DAF), was isolated using the CTAB extraction protocol. The enrichment of mRNA was subsequently achieved with the aid of magnetic oligo (dT) beads. Transcriptome libraries for sequencing were constructed for four sample groups: control peel (ZM_CK), peel treated with 25 mg/L GA₃ (ZM_GA25), peel treated with 50 mg/L GA₃ (ZM_GA50), and peel treated with 75 mg/L GA₃ (ZM_GA75). These RNA-seq libraries were sequenced on the Illumina HiSeq platform, and reads were generated in a 150-bp paired-end format. The reads were quality-filtered using the fastp toolkit that involved the removal of adapter sequences and low-quality reads containing ploy-N [64]. The quality-filtered reads were mapped to the reference cDNA sequences of *Citrus clementina* from the citrus genome annotation project (*Citrus clementina* v1.0) database using the default parameters of HISAT2 [65]. To estimate gene expression levels, fragments per kilobase per million mapped reads (FPKM) of each gene were calculated [66]. Since the sequenced genome was *Citrus clementina* v1.0, the *CsGRFs* were blasted against the *Citrus clementina* v1.0 genome.

Analysis of gene expression by RT-qPCR

Total RNA from ‘Zaomi’ peel was extracted using a plant RNA extraction kit (SteadyPure, ACCURATE BIOLOGY, Changsha, China). Reverse transcription was performed with a reverse transcription kit (Evo M-MLV, ACCURATE BIOLOGY, Changsha, China). Citrus Actin (*Csactin*) was used as the reference gene for normalization in the qPCR analysis. qPCR was performed on the CFX96™ real-time system (Bio-Rad Laboratories). The relative quantification of the amplified products was calculated using the 2^{-ΔΔCT} method [67]. Statistical significance of the qPCR data was assessed using the SPSS 26, based on three independent biological replicates. The primers used for amplification of *CsGRFs* and the *Csactin* was detailed in the supplementary table (Table S7).

The protein association network predictions

The web tool STRING 12.0 (<http://string-db.org>) was used to predict the functional interaction network models of homologs of *CsGRF* proteins in *Arabidopsis thaliana*, and the results were visualized using Cytoscape software.

Abbreviations

GRFs	Growth-regulating factors
GAs	Gibberellins
Cs	Citrus sinensis

Cc	Citrus clementina
Cg	Citrus grandis
DAF	Days after full bloom

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12870-025-06209-2>.

Supplementary Material 1: Supplementary fig. S1: Position of the citrus *GRF* genes on the chromosome. Supplementary fig. S2: Effects of gibberellin on ‘Zaomi’ fruit maximum width. Supplementary fig. S3: Effects of GA₃ on ‘Zaomi’ fruit at the late expansion stage. Supplementary fig. S4: Effects of GA₃ on ‘Zaomi’ fruit at maturation stage. Supplementary table. S1: Detailed information for all identified citrus GRFs. Supplementary table. S2: The collinear correspondence. Supplementary table. S3: Protein blast results of *CsGRFs* with SIGRF4. Supplementary table. S4: Protein BLAST and identity analysis. Supplementary table. S5: Corresponding blast gene mappings. Supplementary table. S6: Expression pattern of *CsGRFs* in leaves ovule, and roots. Supplementary table. S7: Primers used in this study

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Author contributions

YWC and JFY designed the research. JFY and XPL modified the manuscript. XL, BY, MP, YSZ, TC, JWL, FFL, and JFY performed the experiments and analyzed the data. BY provided experimental materials. XL wrote the original manuscript.

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Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request. Gene expression profile data can be obtained at supplementary information. Fruit tissue expression profile data is available on the NCBI Gene Expression Omnibus (GEO) under the accession number GSE125726 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE125726>).

Declarations

Ethics approval and consent to participate

The experimental research on plants performed in this study complies with institutional, national and international guidelines.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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